VLDB 2026 BOSTON

Specifying and Analyzing Transactional Consistency Models with Predicates

Hengfeng Wei Si Liu Yuxing Chen

Hunan University

Texas A&M University

Renmin University of China

September 4, 2026

A semantic and graph-theoretic foundation for predicate operations

Item Operations vs. Predicate Operations

$\mathbf{Read}(x, v)$

$\mathbf{Write}(x, v)$

point access

returns one value

Employee

EID	Salary
X	8
Y	10
Z	6

$\mathbf{PredRead}(P, M)$

$P: \ \mathit{Salary} < 9$

condition-based access

returns a set M of **matched** values

Review

What we know about transactional consistency models

without predicates

Review: Intuitive or Operational Specs

Eswaran et al., 1976

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray, R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

SER

serial execution

Berenson et al., 1995

A Critique of ANSI SQL Isolation Levels

Hal Berenson	Microsoft Corp.	haroldb@microsoft.com
Phil Bernstein	Microsoft Corp.	philbe@microsoft.com
Jim Gray	U.C. Berkeley	gray@crl.com
Jim Melton	Sybase Corp.	jim.melton@sybase.com
Elizabeth O'Neil	UMass/Boston	eoneil@cs.umb.edu
Patrick O'Neil	UMass/Boston	poneil@cs.umb.edu

SI

snapshot + write-conflict control

Review: Axiomatic Specs

Cerone, Bernardi, and Gotsman, 2015

A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

(VIS, AR)

Crooks et al., 2017

Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks
The University of Texas at Austin and Cornell University

Youer Pu
Cornell University

Lorenzo Alvisi
The University of Texas at Austin and Cornell University

Allen Clement
Google, Inc.

Client-Centric Views

Complexity:OOPSLA2019

On the Complexity of Checking Transactional Consistency

RANADEEP BISWAS, Universite de Paris, IRIF, CNRS, France
CONSTANTIN ENEA, Universite de Paris, IRIF, CNRS, France

Checking Complexity

Review: Dependency Graphs

Adya, 1999

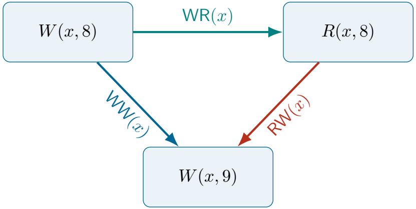
Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science in partial fulfillment of the requirements for the degree of

Doctor of Philosophy



Review: SER Characterization Theorem

Adya, 1999

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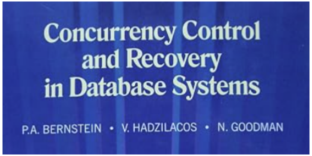
Bernstein, Hadzilacos, and Goodman, 1987



Theorem (roughly stated)

An item-operation history is serializable iff its dependency graph is acyclic.

Review: SI Characterization Theorem

Cerone and Gotsman, 2018

Analysing Snapshot Isolation

ANDREA CERONE, Imperial College London, UK
ALEXEY GOTSMAN, IMDEA Software Institute, Spain

Theorem (roughly stated)

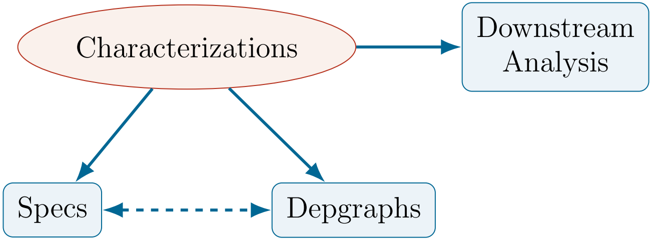
An item-operation history satisfies SI iff every cycle of its dependency graph has at least two adjacent RW *edges.*

Review: Downstream Analysis Tasks

Task	Representative work
Black-box checking	Kingsbury and Alvaro, 2020; Tan et al., 2020; Huang et al., 2023; Cai et al., 2025
CC protocol verification	Xiong, 2020
Robustness	Bernardi and Gotsman, 2016

Review (Summary)

What we know about transactional consistency models
without predicates



Review

What we know about transactional consistency models

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Eswaran et al., 1976

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SI

snapshot + write-conflict control

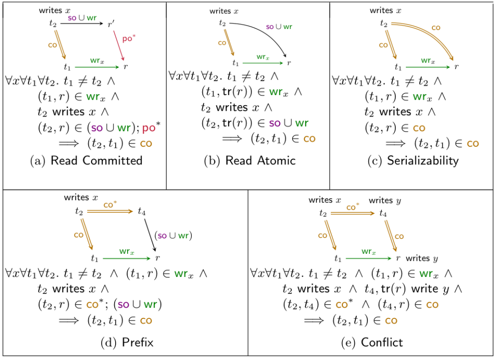
Review: Axiomatic Specs

Bouajjani, Enea, and Román-Calvo, 2025

On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

Ahmed Bouajjani[1], Constantin Enea[2], and Enrique Román-Calvo[1]

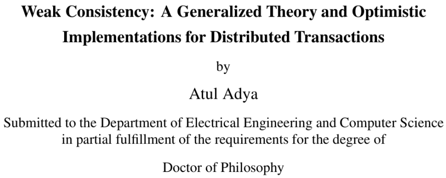
[1] Université Paris Cité, CNRS, IRIF
{abou, calvo}@irif.fr
[2] LIX, École Polytechnique, CNRS and Institut Polytechnique de Paris
cenea@lix.polytechnique.fr



On checking complexity and algorithms

Review: Dependency Graphs

Adya, 1999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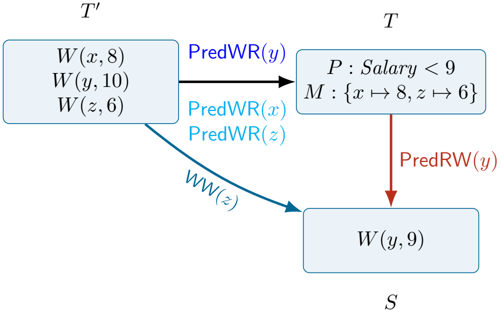
T also "observes" **unmatched** $W(y, 10)$ of T'



S later than T' **changes matches** of T on y

Review: Dependency Graphs

Predicate-read-dependency, $T \xrightarrow{\mathsf{PredWR}(x)} S$, definitions differ.

Predicate-anti-dependency, $T \xrightarrow{\mathsf{PredRW}(x)} S$, definitions differ.

Adya, 1999

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S changes matches

Sun and Zou, 2025

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

the "earliest" S

Complexity:CAV2025

On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

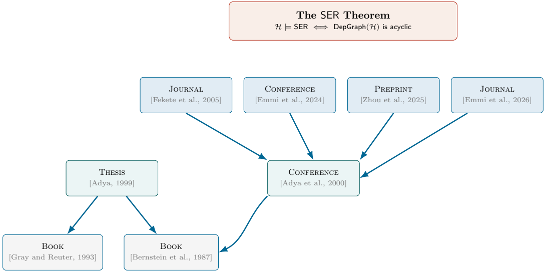
Ahmed Bouajjani[1], Constantin Enea[2], and Enrique Román-Calvo[1]

[1] Université Paris Cité, CNRS, IRIF
{abou, calvo}@irif.fr
[2] LIX, École Polytechnique, CNRS and Institut Polytechnique de Paris
cenea@lix.polytechnique.fr

CAV Artifact Evaluation ★ Available

no "change matches"

Review: SER Characterization



3.4 Correctness and Flexibility of the New Specifications

Our conditions for PL-3 provide the well-known notion of conflict-serializability [BHG87, GR93]. That is, DSG(H) is acyclic, and G1a, and G1b are satisfied for a history H iff H is conflict-serializable. We can prove the equivalence of our PL-3 conditions with conflict-serializability using the proof given in [GR93]; our DSGs are similar to their graphs. This equivalence can also be proved along the lines of the proof given in [BHG87] for view-serializability.

No rigorous proof in the literature

Review: SI Characterization

Adya, 1999

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Theorem 2: A history H consisting of committed transactions executes under Snapshot Isolation iff G1 and G-SI are disallowed.

G-SIa: Low-level *start-depends* $\prec_t$ relation

Definition 17 : **Snapshot Read.** All reads performed by a transaction T_i occur at its start point. That is, if $r_i(x_j)$ occurs in history H, then:

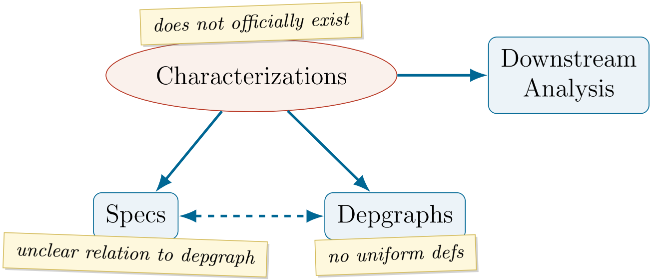
1. $c_j \prec_t s_i$, and
2. if $w_k(x_k)$ also occurs in H ($j \neq k$), then either
   (a) $s_i \prec_t c_k$, or
   (b) $c_k \prec_t s_i$ and $x_k \ll x_j$

Definition 18 : **Snapshot Write.** If T_i and T_j are concurrent, they cannot both modify the same object. That is, if $w_i(x_i)$ and $w_j(x_j)$ both occur in history H, then either $c_i \prec_t s_j$ or $c_j \prec_t s_i$. This is the first-committer-wins property.

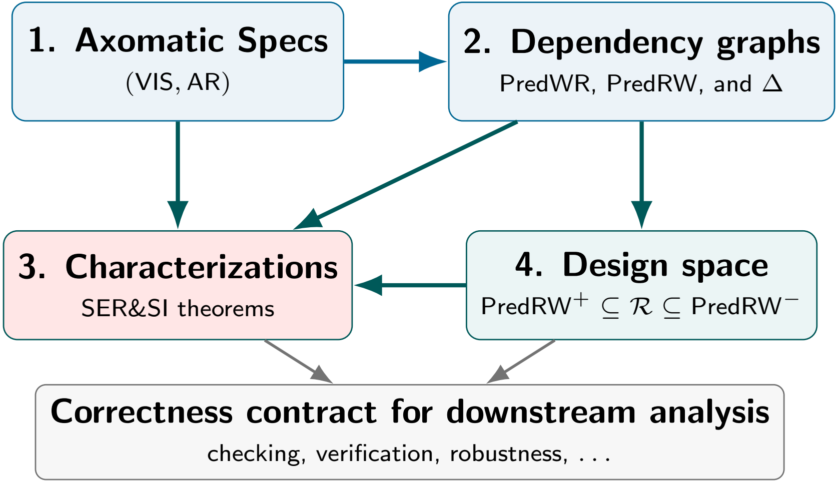
Proof based on *item-only* properties of SI

Review (Summary)

What we know about transactional consistency models
with predicates



Contributions: Fill and Exploit the Gap



1: Axiomatic Specs

Table 2: Consistency axioms, constraining an abstract execution $(\mathcal{H}, \mathsf{VIS}, \mathsf{AR})$. (Generalized based on [11])

$$\forall (\mathsf{E}, \mathsf{po}) \in \mathcal{T}, e \in \mathsf{E}, x \in \mathsf{K}, v \in \mathsf{V}, F \subseteq \mathsf{E}.$$
$$\Big(\mathsf{op}(e) = \mathsf{R}(x,v) \wedge F = \{f \mid (\mathsf{op}(f) = \mathsf{R}(x,\_) \vee \mathsf{op}(f) = \mathsf{W}(x,\_)) \wedge f \xrightarrow{\mathsf{po}} e\} \wedge F \neq \emptyset\Big)$$
$$\implies \mathsf{op}\big(\max_{\mathsf{po}} F\big) = \_(x,v) \qquad \text{(INT-ITEM)}$$

$$\forall (\mathsf{E}, \mathsf{po}) \in \mathcal{T}, e \in \mathsf{E}, f \in \mathsf{E}, x \in \mathsf{K}, v \in \mathsf{V}, P \in \mathsf{P}, M \subseteq \mathsf{K} \times \mathsf{V}, M' \subseteq \mathsf{K} \times \mathsf{V}, F \subseteq \mathsf{E}, G \subseteq \mathsf{E}.$$
$$\Big(\mathsf{op}(e) = \mathsf{PR}(P,M) \wedge \mathsf{op}(f) = \mathsf{PR}(P,M') \wedge f \xrightarrow{\mathsf{po}} e \wedge \neg(\exists g \in \mathsf{E}.\ \mathsf{op}(g) = \mathsf{PR}(P,\_) \wedge f \xrightarrow{\mathsf{po}} g \xrightarrow{\mathsf{po}} e) \wedge$$
$$F = \{h \mid \mathsf{op}(h) = \mathsf{W}(x,\_) \wedge f \xrightarrow{\mathsf{po}} h \xrightarrow{\mathsf{po}} e\}\Big)$$
$$\implies \Big((F = \emptyset \implies ((x,v) \in M \iff (x,v) \in M')) \wedge$$
$$(F \neq \emptyset \implies ((x,v) \in M \iff (\mathsf{op}(\max_{\mathsf{po}} F) = \mathsf{W}(x,v) \wedge P(x,v))))\Big) \wedge \qquad \text{(INT-PRED)}$$
$$\Big(\mathsf{op}(e) = \mathsf{PR}(P,M) \wedge \neg(\exists g \in \mathsf{E}.\ \mathsf{op}(g) = \mathsf{PR}(P,\_) \wedge g \xrightarrow{\mathsf{po}} e) \wedge$$
$$G = \{h \mid \mathsf{op}(h) = \mathsf{W}(x,\_) \wedge h \xrightarrow{\mathsf{po}} e\} \wedge G \neq \emptyset\Big)$$
$$\implies ((x,v) \in M \iff (\mathsf{op}(\max_{\mathsf{po}} G) = \mathsf{W}(x,v) \wedge P(x,v)))$$

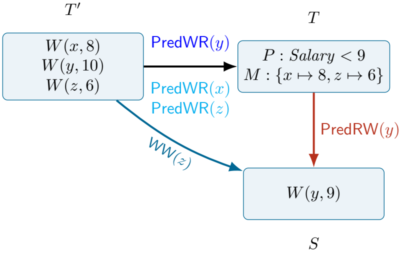
$$\forall T \in \mathcal{T}, x \in \mathsf{K}, v \in \mathsf{V}.\ T \vdash \mathsf{R}(x,v) \implies \max_{\mathsf{AR}}\big(\mathsf{VIS}^{-1}(T) \cap \mathsf{WriteTx}_x\big) \vdash \mathsf{W}(x,v) \qquad \text{(EXT-ITEM)}$$

$$\mathsf{SO} \subseteq \mathsf{VIS} \qquad \text{(SESSION)}$$

$$\forall T \in \mathcal{T}, P \in \mathsf{P}, M \subseteq \mathsf{K} \times \mathsf{V}, x \in \mathsf{K}.\ T \vdash \mathsf{PR}(P,M,x) \implies$$
$$\Big(\boxed{x \in \mathrm{dom}(M)} \implies \exists v \in \mathsf{V}.\ (x,v) \in M \wedge \max_{\mathsf{AR}}\big(\mathsf{VIS}^{-1}(T) \cap \mathsf{WriteTx}_x\big) \vdash \mathsf{W}(x,v)\Big) \wedge$$
$$\Big(\boxed{x \in \mathsf{K} \setminus \mathrm{dom}(M)} \implies \forall v \in \mathsf{V}.\ \big(\max_{\mathsf{AR}}\big(\mathsf{VIS}^{-1}(T) \cap \mathsf{WriteTx}_x\big) \vdash \mathsf{W}(x,v) \implies \neg P(x,v)\big)\Big) \qquad \boxed{\text{(EXT-PRED)}}$$

$$\mathsf{AR}\ ;\ \mathsf{VIS} \subseteq \mathsf{VIS} \quad \text{(PREFIX)} \qquad \mathsf{VIS} = \mathsf{AR} \quad \text{(TOTALVIS)}$$

$$\forall S, T \in \mathcal{H}.\ S \bowtie T \implies (S \xrightarrow{\mathsf{VIS}} T \vee T \xrightarrow{\mathsf{VIS}} S) \qquad \text{(NOCONFLICT)}$$



External predicate consistency (EXT-PRED axiom)

For each externally observed key x, choose the AR-latest writer visible to the reader.

Its value must equal the returned value when x **appears** in the result,

and must fail the predicate when x is **omitted**.

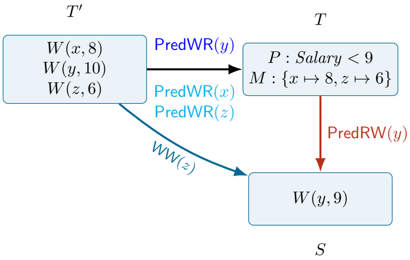
2: Dependency Graphs

Graph witness

$T' \xrightarrow{\text{PredWR}(y)} T$: T' supplies the version by which T externally observes x.

Observation-changing successor

$T \xrightarrow{\text{PredRW}(y)} S$: $T' \xrightarrow{\text{WW}(y)} S$ and S **changes the matches** of T on y wrt T'.



3: Characterizations

$$\mathrm{PredRW}^{+} \subseteq \mathcal{R} \subseteq \mathrm{PredRW}^{-}$$

Lower frontier

The earliest later write that changes the observation. It is enough to expose a predicate-consistency violation.

Baseline

All observation-changing later writers. This is the direct "changes matches" reading.

Upper closure

All later writers sharing a PredWR/WW witness. It is safe to orient in a witnessing execution.

Different fixed graphs; the same existential SER/SI history characterization. The theorem separates correctness from representation.

Adya, 1999; Sun and Zou, 2025; Bouajjani, Enea, and Román-Calvo, 2025

4: Design Space (of Predicate Anti-Dependencies)

$$\mathsf{PredRW}^{+} \subseteq \mathcal{R} \subseteq \mathsf{PredRW}^{-}$$

Lower frontier

The earliest later write that changes the observation. It is enough to expose a predicate-consistency violation.

Baseline

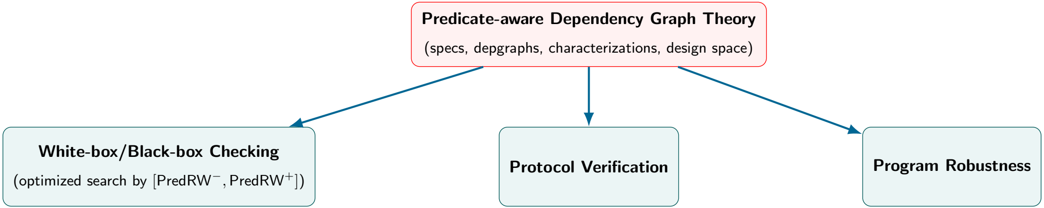
All observation-changing later writers. This is the direct "changes matches" reading.

Upper closure

All later writers sharing a PredWR/WW witness. It is safe to orient in a witnessing execution.

The same existential SER/SI history characterization.

Future Work: Downstream Analysis



[PredRW^-, PredRW^+]-guided checker

- a lower persistent cycle can reject;
- an upper acyclic graph can accept;
- ambiguous cases are refined or solved

Takeaways: Theory Established, Downstrem Analysis Enabled

1. Specs. Extend $(\mathsf{VIS}, \mathsf{AR})$ with item-wise external predicate observations.
2. Dependency graphs. Formally define PredRW with the "change matches" condition.
3. SER/SI Characterizations. Prove the same execution–graph equivalence style as the item-only results, while resolving the predicate-specific SI witness ambiguity.
4. Proof-theoretic design space. Every relation $\mathsf{PredRW}^{-} \subseteq \mathcal{R} \subseteq \mathsf{PredRW}^{+}$ preserves the same existential history-level characterization.

Checking, Verification, Robustness, ...